Need of set operations

1. No duplicates in result set
2. Distinct rows

Joins	Set Operations
• No need of same attributes/ types	• Same attributes on both tables
• Conditions met	• Set operations
• Distinct Columns	• Distinct rows

Set Operations on MySQL

1. Union / Union ALL
2. Intersection
3. Minus

T_1 UNION T_2

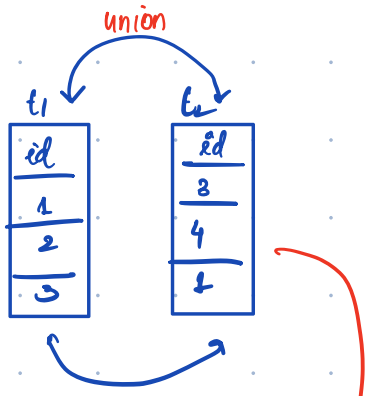
UNION OPERATION

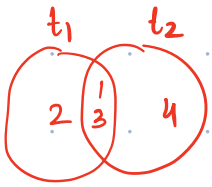
SELECT * from t_1

Union

Select * from t_2;

Number of Columns } t_1 & t_2
Order of Column } (follow same)





= 4 elements

Result

id
1
2
3
4

CREATE TABLE t1 (id INT);

CREATE TABLE t2 (id INT);

INSERT INTO t_1 (id) VALUES ('1'),('2'),('3');

INSERT INTO t_2 (id) VALUES ('1'),('4'),('3');

SELECT * from t_1;

SELECT * from t_2;

SELECT * FROM t_1 UNION SELECT * FROM t_2;

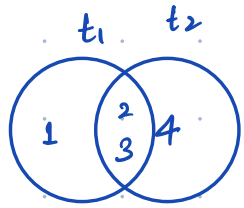
-UNION ALL (we'll get duplicate values)

[NOTE : No. of columns should be same during Union operation in both tables → SELECT * FROM t_1 UNION SELECT id FROM t_2] throws error.

INTERSECT

$t_1 = (1,2,3)$

$t_2 = (2,3,4)$

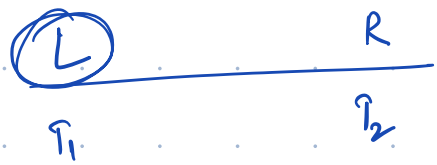


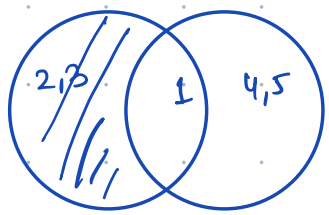

```
SELECT * FROM t1
INTERSECT
SELECT * FROM t2
```

This keyword not available in mySQL

```
SELECT DISTINCT id
FROM t1 INNER JOIN t2 using (id);
```

MINUS OPERATION





$T_1 - T_2$

$T_1 = \{1, 2, 3\}$

$T_2 = \{1, 4, 5\}$

$\longrightarrow T_1 - T_2 = \{2, 3\}$

```
SELECT id from T1
    LEFT JOIN T2 using (id)
    WHERE t2.id is NULL;
```

1, 2, 3

Note: All the set operators have same order of precedence